

Date	02 July 2026
Team ID	XXX
Project Name	Customer Persona Generator - AI Powered Marketing Persona Application
Maximum Marks	1 Mark

SKILL WALLET

A SmartBridge Product

SCALABILITY & FUTURE PLAN TEMPLATE

Description: Describe how the current solution can scale to handle increased load or users, and outline the features or enhancements planned for future releases.

SCALABILITY

Aspect	Details
Current Architecture	A modular microservices architecture deployed on cloud infrastructure. It utilizes a responsive web-based frontend, a lightweight Python (FastAPI) backend framework for processing logic, and asynchronous task managers to queue and handle requests directed to Generative AI foundation models.
Scalability Approach	Horizontal auto-scaling via containerized deployment (Docker & Kubernetes) to handle concurrent traffic spikes during peak marketing campaign seasons. Implements a robust caching layer (Redis) to store common industry segment frameworks, minimizing redundant AI model calls and optimizing database load.
Expected Growth / Load	Initial pilot phase scales for 500 active marketing users. Anticipating a rapid multi-fold expansion to over 15,000 monthly active users within 6 months, scaling processing demand from 2,000 to over 50,000 distinct multi-segment persona generations per week.

FUTURE PLAN

S.No	Planned Feature / Enhancement	Target Timeline	Priority
1	Direct CRM & Marketing Automation Sync: Seamless API integration to export generated personas directly into platforms like HubSpot, Salesforce, and Marketo.	August 2026	High
2	Multi-Modal Persona Profiles: Enhancing profiles with AI-generated target consumer avatars, synthetic dynamic voice bios, and sample localized ad copy variations.	October 2026	Medium
3	Real-Time Social Trend Scraping: Integrating live social media and market search data feeds to dynamically update persona behavior patterns based on emerging market shifts.	December 2026	High
4	A/B Testing Simulation Engine: A predictive module to run virtual simulation campaigns against generated personas to project click-through rates (CTR) and ROI.	February 2027	Low